

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

A scientist measured the lengths of **240** gray seals from Muskeget Island and **120** gray seals from Sable Island. The scientist determined that the mean length of the **240** gray seals from Muskeget Island was **88** inches and the mean length of the **120** gray seals from Sable Island was **94** inches. What was the mean length of all **360** gray seals the scientist measured for this study?

Answer

- A. **89**
- B. **90**
- C. **91**
- D. **92**

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is calculated by dividing the sum of the values in the data set by the number of values in the data set. It's given that the mean length of the **240** gray seals from Muskeget Island was **88** inches. This can be represented by the equation $\frac{x}{240} = 88$, where x represents the sum of the lengths, in inches, of the **240** gray seals from Muskeget Island. Multiplying both sides of this equation by **240** yields $x = (240)(88)$, or $x = 21,120$ inches. It's also given that the mean length of the **120** gray seals from Sable Island was **94** inches. This can be represented by the equation $\frac{y}{120} = 94$, where y represents the sum of the lengths, in inches, of the **120** gray seals from Sable Island. Multiplying both sides of this equation by **120** yields $y = (120)(94)$, or $y = 11,280$ inches. Therefore, the sum of the lengths, in inches, of all **360** gray seals is **21,120 + 11,280**, or **32,400**. Dividing this sum by **360** yields $\frac{32,400}{360}$, or **90** inches. Therefore, the mean length of all **360** gray seals the scientist measured for this study is **90** inches.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the mean length of **88** inches and **94** inches, not the mean length of all **360** gray seals the scientist measured for this study.

Choice D is incorrect and may result from conceptual or calculation errors.